Beyond Reduced-Order Control: Learning the Coupled Nine-DOF Nonlinear Dynamics of a Wave Energy Converter

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Honours research with WaveX

THE UNIVERSITY OF WESTERN AUSTRALIA · WaveX · NEURAL INFORMATION PROCESSING SYSTEMS

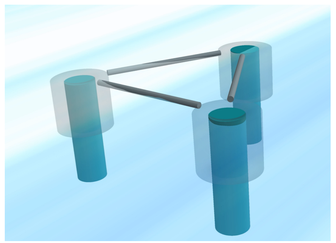
CSSE 50th anniversary

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THE CONTROL CHALLENGE

How hard should we resist the next wave?

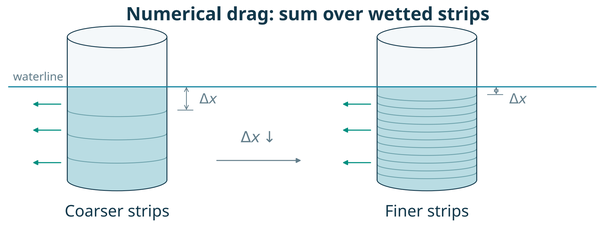
Ocean waves drive each D-Spar's relative motion. PTO damping sets the resistance and changes the resulting motion.



Each inner spar heaves independently relative to its outer spar. The rigid frame couples the platform: 6 platform DOFs + 3 relative-heave DOFs. Water-surface colour shows ocean-wave elevation.

CALCULATE THE KNOWN PHYSICS; LEARN THE EXPENSIVE NONLINEAR CORRECTION

Grey-box recursive MLP rollout



$$F_D \approx \sum_s -\tfrac{1}{2}\rho C_D D_s \Delta x_s u_s |u_s|$$

Smaller Δx: more strips, finer force-integration resolution

Keep known mechanics; learn the nonlinear correction

$$(\mathbf{M} + \mathbf{A}_\infty)\dot{\mathbf{v}} = \mathbf{F}_{\text{known}} + \widehat{\boldsymbol{\phi}}_\theta(\mathbf{q}, \mathbf{v})$$

Known: ocean-wave excitation, radiation memory, mechanical losses, PTO damping and linear restoring.

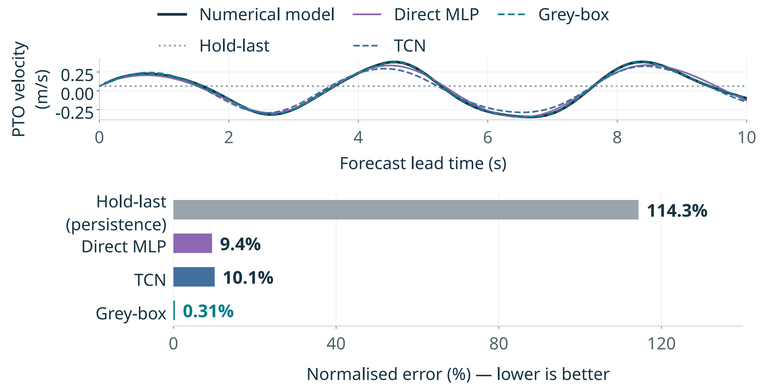
$$\boldsymbol{\phi} = \mathbf{F}_{\text{drag}} + \mathbf{C}\mathbf{q} - \mathbf{F}_{\text{rest, nl}}$$

Nonlinear body drag + geometry-dependent buoyancy/gravity

$$(\mathbf{q}, \mathbf{v})_t \longrightarrow \widehat{\boldsymbol{\phi}}_\theta \longrightarrow \text{RK4} \longrightarrow (\mathbf{q}, \mathbf{v})_{t+\Delta t}$$

PREDICTING MOTION

10 s forecasts of 9 displacements and 9 velocities.



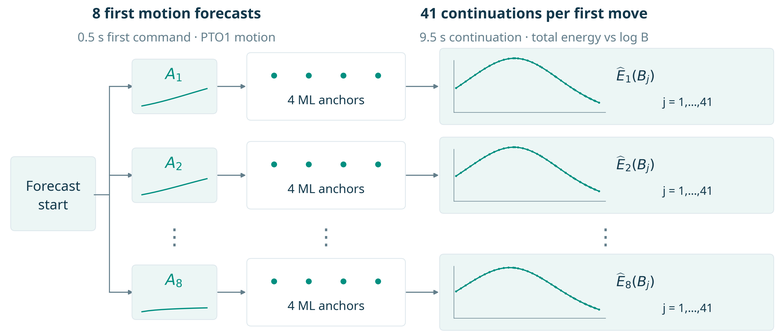
100% = RMSE equal to the true-signal standard deviation.

Fine-tuned Chronos-2 — **23.28%** NRMSE
Changing-damping validation
177 forecasts · one wave · 10 s horizon

Main comparison: 2 waves × 8 static damping settings.
Earlier frozen models; native training/input budgets differ.

NONLINEAR PREDICTIVE CONTROL

Explore 8 × 41 candidate trajectories



8 × 41 = 328 candidate control sequences

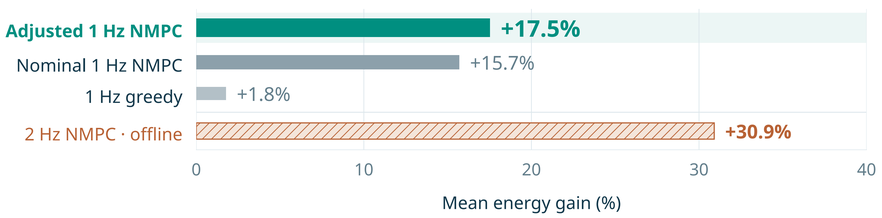
Archived model predictions · fitting example · shared motion / energy scales.

$$\max_{i,j} E_{ij} = \int_0^{10\,\text{s}} B_{ij}(t) \sum_{k=7}^{9} v_k^2(t)\, dt$$

Adjusted 1 Hz: execute A then 0.88B, 0.5 s each (bounded).
2 Hz: execute A for 0.5 s, then replan. See QR for full settings.

RESULTS IN NUMERICAL SIMULATION

+17.5% mean mechanical energy adjusted 1 Hz NMPC



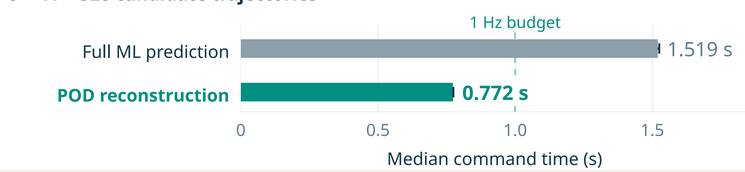
2 Hz is offline: 0.77 s median computation exceeds its 0.5 s budget.

Mean paired gain over fixed B = 5,000 N·s/m; 9 states × 4 waves.
300 s per wave; known future waves; combined descriptive cohort (original two + fresh two waves per state).

Full ML prediction and POD

Reference nonlinear model shown separately

8 × 41 = 328 candidate trajectories



Reference NL numerical model: 40.84 s median
Separate earlier search: 41 screen actions + 42–49 plans · CPU / MATLAB

Current: 8 requests × 10; same initial POD screen; RTX 5000, 4 lanes.
Native: 8 requests × 3, CPU/MATLAB. Both: Hs = 0.75 m, Tp = 3 s.

Explore the project

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Paper, results & contact

reuben-trade.github.io/wavex/

Earlier CEED paper and updated results are clearly identified on the project page.

Scope: known ocean-wave preview; uniform PTO damping; 1 s application delay for both controllers; numerical validation. Next: causal forecasts and device trials. Mentors: Tim Green and Amal Jose Arackal.